CONCEPT NOTE

Predicting Maternal Health Risk Among Pregnant Women in Kenya

Using Machine Learning and Bayesian Inference on the Kenya Demographic and Health Survey 2022

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🚨 Problem Statement

Kenya's maternal mortality rate stands at 355 deaths per 100,000 live births — most preventable. Community Health Workers (CHWs) serve over 100,000 households but rely on paper checklists with no data-driven tool to identify which pregnant women are truly at risk before complications arise.

No deployed, individual-level, probabilistic maternal risk scoring tool exists for CHWs in Kenya using nationally representative data.

🎯 Objectives

1. Build a machine learning classification model that predicts individual maternal health risk (Low / Moderate / High) using KDHS 2022 data.

2. Integrate Bayesian inference to produce probabilistic risk scores with uncertainty quantification, enabling more trustworthy clinical decisions.

3. Develop a Claude AI-powered web application that translates model outputs into plain-language explanations, care recommendations, and referral letters for CHWs in English and Swahili.

🔧 Methodology

Data: KDHS 2022 Individual Recode — 10,391 women with complete maternal records, 13 variables selected.

Preprocessing: Imputation, encoding, scaling, SMOTE for class imbalance.

Models: Logistic Regression (baseline), Random Forest, XGBoost, CatBoost, Stacking Ensemble with Optuna tuning.

Bayesian: PyMC for posterior probability distributions and credible intervals per prediction.

Explainability: SHAP values for global and individual-level feature importance.

End Product: Streamlit web app + Claude AI for risk explanation, recommendations, and referral letter generation.

💡 Expected Impact

Clinical: CHWs can identify and refer high-risk pregnant women before complications occur, reducing preventable maternal deaths.

Policy: County-level risk dashboard enables MOH and NGOs to allocate resources to highest-burden areas across Kenya's 47 counties.

Academic: First individual-level, Bayesian ML maternal risk tool built on KDHS 2022 data — fills a confirmed research gap in Kenyan health informatics.

SDG Alignment: Directly contributes to UN SDG 3 (Good Health & Well-being) and Kenya's Universal Health Coverage agenda.

AI Innovation: Integration of generative AI (Claude) into a CHW decision support tool is a novel contribution not present in any reviewed literature.